

Hence $y_2(x) = x^{-1/2} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right)$, when $a_0 = 1$

$$= \frac{1}{\sqrt{x}} \cos x \text{ is second solution.}$$

$\therefore$ The general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= \frac{1}{\sqrt{x}} (c_1 \sin x + c_2 \cos x) \text{ (Ans.)}$$

Example-4 Solve Legendre's differential equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$, when n is a constant.

Solution Given differential equation is

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \dots(1)$$

Let the solution of (1) be

$$y = \sum_{r=0}^{\infty} a_r x^{c+r}, a_0 \neq 0 \quad \dots (2)$$

$$\therefore y' = \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1}$$

$$\text{and } y'' = \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2}$$

Now putting the values of y, y', y'' in (1), we get

$$(1-x^2) \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - 2x \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1} + n(n+1) \sum_{r=0}^{\infty} a_r x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} 2a_r (c+r) x^{c+r-1} - \sum_{r=0}^{\infty} 2a_r (c+r) x^{c+r} + (n^2 + n) \sum_{r=0}^{\infty} a_r x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r [(c+r)^2 - (c+r) + 2(c+r) - n^2 - n] x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r [(c+r)^2 + (c+r) - n(n+1)] x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1) x^{c-2} + a_1 (c+1) c x^{c-1} + \sum_{r=2}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1) x^{c-2} + a_1 (c+1) c x^{c-1} + \sum_{r=0}^{\infty} a_{r+2} (c+r+2)(c+r+1) x^{c+r} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1)x^{c-2} + a_1(c+1)cx^{c-1} + \sum_{r=0}^{\infty} \{a_{r+2}(c+r+2)(c+r+1) - a_r(c+r-n)(c+r+n+1)\}x^{c+r} = 0$$

Equating to zero the coefficient of lowest power of x (coefficient of x^{c-2}), then the indicial equation is

$$a_0 c(c-1) = 0$$

$$\Rightarrow c(c-1) = 0 \quad [\because a_0 \neq 0]$$

$$\therefore c = 0, 1$$

Also equating to zero the coefficient of x^{c-1} , we get

$$a_1(c+1)c = 0$$

$$\Rightarrow a_1 = \frac{0}{c(c+1)} \quad [\because a_0 \neq 0]$$

$$\text{For } c = 0, \quad a_1 = \frac{0}{0} \quad [\text{Indeterminate form}]$$

Again equating to zero the coefficient of x^{c+r} , then the recurrence relation is

$$a_{r+2}(c+r+2)(c+r+1) - a_r(c+r-n)(c+r+n+1) = 0$$

$$\Rightarrow a_{r+2} = \frac{(c+r-n)(c+r+n+1)}{(c+r+2)(c+r+1)} a_r \quad \dots (3)$$

When $c = 0$, the recurrence relation (3) becomes

$$\Rightarrow a_{r+2} = \frac{(r-n)(r+n+1)}{(r+1)(r+2)} a_r \quad \dots (4)$$

Now putting $r = 0, 1, 2, 3, 4, 5, 6, \dots$ in (4), we get

$$a_2 = -\frac{n(n+1)}{1 \cdot 2} a_0$$

$$a_3 = -\frac{(n-1)(n+2)}{2 \cdot 3} a_1$$

$$a_4 = \frac{(2-n)(n+3)}{3 \cdot 4} a_2 = \frac{(n-2)n(n+1)(n+3)}{1 \cdot 2 \cdot 3 \cdot 4} a_0$$

$$a_5 = \frac{(3-n)(n+4)}{4 \cdot 5} a_3 = \frac{(n-3)(n-1)(n+2)(n+4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} a_1$$

and so on.

From (2), we get

$$y = x^0(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots)$$

$$\Rightarrow y = x^0 \left[a_0 + a_1x - \frac{n(n+1)}{1 \cdot 2} a_0x^2 - \frac{(n-1)(n+2)}{2 \cdot 3} a_1x^3 + \frac{(n-2)n(n+1)(n+3)}{1 \cdot 2 \cdot 3 \cdot 4} a_0x^4 \right. \\ \left. + \frac{(n-3)(n-1)(n+2)(n+4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} a_1x^5 + \dots \right]$$

$$\Rightarrow y = a_0 \left[1 - \frac{n(n+1)}{2!} x^2 + \frac{(n-2)n(n+1)(n+3)}{4!} x^4 - \dots \right] \\ + a_1 \left[x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 - \dots \right]$$

$\Rightarrow y = Au(x) + Bv(x)$, where A, B are two arbitrary constants and $u(x), v(x)$

represents two solutions.

Example-5 Solve Hermite differential equation $\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2ny = 0$, where n is a constant.

Solution Try yourself.

Example-6 Solve Laguerre differential equation $x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} + ny = 0$, where n is a non-negative integer.

Solution Try yourself.

The End